Solver Documentation
electroChem

Thor Dremstrup

AARHUS UNIVERSITY

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Contents

1 Introduction

The purpose of this document is to describe the custom CFD solver for electrochemical systems explaining the physical interpretation and numerical implementation. The model is applied to microbial electrosynthesis (MES) of acetate via insitu production of hydrogen and external supply of carbon dioxide.

1.1 System

The system is a thin flow-through electrolytic cell that is supplied water with dissolved carbon-dioxide and buffer ions. The reactions in the system happens on multiple timescales, electrode and microbial reactions occur on the transport timescale and are treated kinetically. However, the water and carbonate acid-base reactions occur on timescales much shorter than transport and are therefore assumed to remain in local thermodynamic equilibrium.

Kinetic reactions

Once current is passed through the cell hydrogen is made at the cathode

$$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^- \tag{1}$$

and oxygen at the anode

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \tag{2}$$

Microbes exists in either a biofilm at the cathode or on planktonic form in the electrolyte where they perform acetegonesis

$$2CO_2 + 4H_2 \rightarrow CH_3COO^- + H^+ + 2H_2O \tag{3}$$

Equilibrium reactions

In the electrolyte ionization of water, carbon, acetate and buffer species follow the equilibrium reactions

$$H_2O \leftrightarrow H^+ + OH^- \tag{4}$$

$$H_2O + CO_2 \leftrightarrow HCO_3^- + H^+ \tag{5}$$

$$HCO_3^- \leftrightarrow CO_3^{2-} + H^+ \tag{6}$$

$$CH_3COOH \leftrightarrow CH_3COO^- + H^+ \tag{7}$$

The acetate speciation reaction is not expected to have any significant impact on the system as the acetic-acid (CH_3COOH) is only present at pH below roughly 4.5. pH is expected to be near neutral in the bulk, locally acetic towards the anode and locally alkaline towards the cathode. Acetate requires hydrogen to be produced and hydrogen is only made at the cathode which is locally alkaline so the acetic-acid form will not be present in the system and the acetate dissociation equilibrium is neglected unless high mixing is observed.

Salt dissociation

The dissociation of sodium is assumed complete owing to its very large dissociation constant combined with the relatively low concentration of buffer salts used in MES systems. This means that the reaction is practically one way leaving only negligible trace amounts of $NaHCO_3$

$$NaHCO_3 \rightarrow Na^+ + HCO_3^- \tag{8}$$

Therefore the buffer ions are treated as completely dissociated in the electrolyte before entering the domain.

System schematic

Based on the above reactions and simplification the complete system is as presented in Figure 1

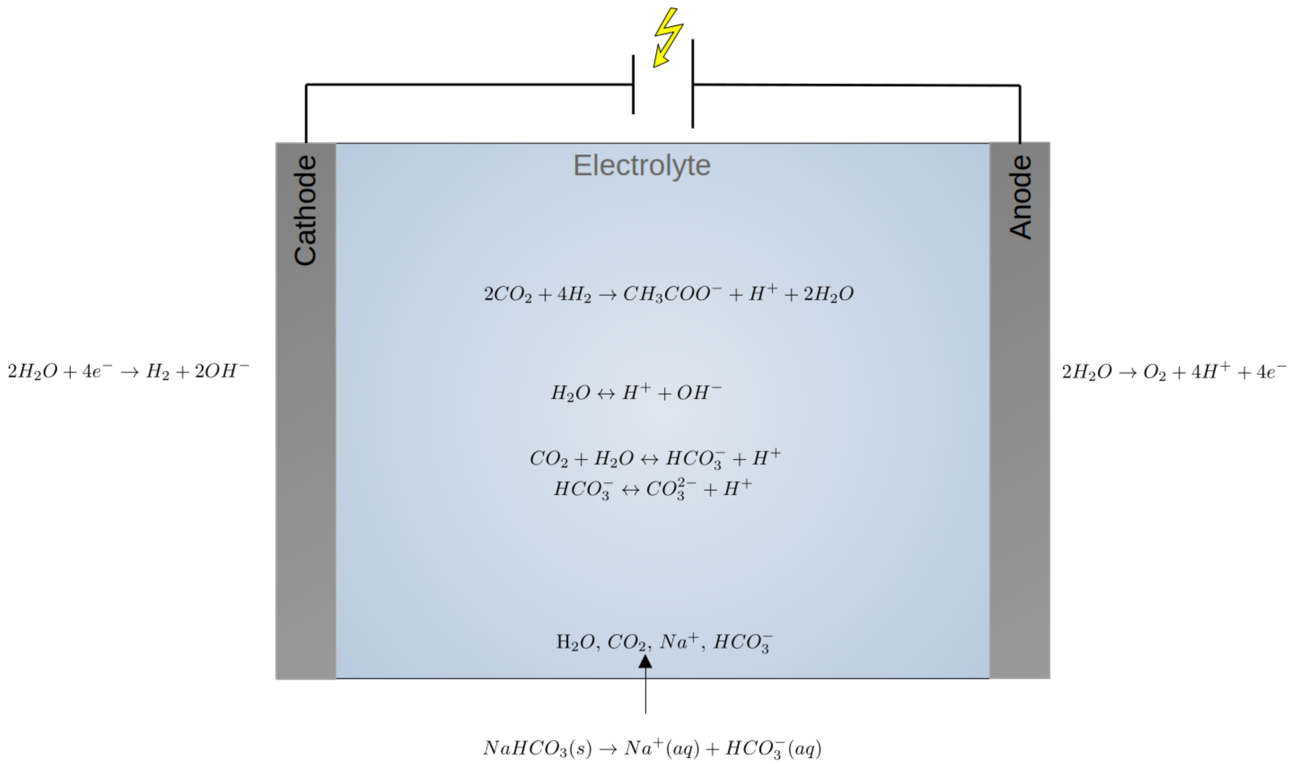

Figure 1: System schematic

1.2 Assumptions

The system is build on the following underlying assumptions

- Incompressible laminar flow
- Isothermal
- Single-phase
- Dilute electrolyte phase

1.3 Conservative Equilibrium Projection

The challenge of this setup is the simple electrolyte composition with very dilute ionic concentrations. This means that even small concentrations of a given species can have a significant impact on the pH. For this reason the equilibrium reactions are important if realistic pH gradients are desired. Existing models already treat multicomponent ionic equilibriums such as Simoni et al. 2005. Their approach creates conserved inventories/ components through transport and based on these lets equilibrium redistribute the species with the constraint of conserving the components. And example is saying that the total inorganic carbon has been transported during a given time-step, then equilibrium can redistribute it to carbon-dioxide, bicarbonate and carbonate to satisfy equilibrium, but the net amount of inorganic carbon must be preserved.
This method gives realistic pH gradients for multicomponent systems based solely on the thermodynamic equilibrium. In electrolytic cells the bulk electrolyte is expected to have no net charge outside of the double layer described bye the Debye-length

$$\lambda_D = \sqrt{\frac{\epsilon RT}{F^2 \sum_i z_i^2 c_i}}, \tag{9}$$

This model will not resolve length-scales approaching the Debye-length so

$$L \gg \lambda_D. \tag{10}$$

And not net charge is expected in the bulk. Under this assumption the bulk electrolyte satisfies

$$\sum_i z_i c_i = 0, \tag{11}$$

There are multiple ways to treat this, Janotta et. al compared different approaches in their work Janotta et al. 2024. The simplest way to enforce electro-neutrality is by choosing a mobile ion that is not solved through transport but explicitly to enforce electro-neutrality. This however can break mass conservation. Rather than eliminating a mobile ion, the electrical potential is obtained from current conservation,

$$\nabla \cdot j = 0 \tag{12}$$

with

$$j = -F \sum_i z_i D_i \nabla c_i - \frac{F^2}{RT} \sum_i z_i^2 D_i c_i \nabla \phi \tag{13}$$

and solve ϕ such that current conservation is satisfied. This reflects the electric field required to transport the electrolyte without charge accumulation. While this formulation correctly describes electrochemical transport, it does not by itself account for the rapid redistribution of species caused by acid-base equilibrium. Conversely, equilibrium calculations alone determine the local chemical composition but contain no information about the electric field required to transport that composition. The present solver therefore couples these two descriptions via Conservative Equilibrium Projection (CEP). The conserved components treatment of equilibrium from Simoni et al. 2005 is combined with electro-neutrality to couple the thermodynamic equilibriums and electrochemistry of the system. This is done by solving the equilibriums together with the electro-neutrality constraint

$$K = \frac{\prod c_R^{\nu}}{\prod c_P^{\nu}} \tag{14}$$

$$\sum_i z_i c_i = 0$$

and then solving ϕ to satisfy current conservation. Thereby utilizing the equilibrium constants to find thermodynamically stable equilibrium states and applying electro-neutrality to chose one that satisfies no net charge and lastly providing the necessary electric field to transport it. Consequently, the thermodynamic and electrochemical constraints are satisfied simultaneously without introducing an artificial transport equation for a selected mobile ion. For this system CEP reduces to a scalar non-linear equation of H^+ together with a iterative current-conservation solve that allows for treating the equilibrium reactions without resolving the forward and backwards kinetics.

2 Numerical implementation

This chapter will cover the numerical implementation of the electroChem solver algorithm. The solver is an extension of the existing OpenFOAM solver pimpleFoam for incompressible flow using the PIMPLE algorithm. Therby the flow is solved via the incompressible Navier-Stokes formulation

$$\nabla \cdot \mathbf{u} = 0 \tag{15}$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho}\nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \tag{16}$$

Then using the resulting velocity the provisional concentrations c^* are found for each species i using the current potential $\phi^{(k)}$

$$\frac{\partial c_i^*}{\partial t} + \nabla \cdot (c_i^* \mathbf{u}) = \nabla \cdot (D_i \nabla c_i^*) + \nabla \left(D_i \frac{z_i F}{RT} c_i^* \nabla \phi^{(k)} \right) + R_i \tag{17}$$

with reaction source

$$R_i = \nu_i \dot{r}_m \tag{18}$$

where $\dot{r}_m$ is the reaction rate of acetegonesis

$$\dot{r}_m = X \dot{q}_{max} \left(\frac{c_{H^+}}{c_{H^+} + K_{S.H^+}} \right) \left(\frac{c_{CO_2}}{c_{CO_2} + K_{S.CO_2}} \right) \tag{19}$$

From the provisional species concentration conserved components are defined as

$$C_T = c^*_{CO_2} + c^*_{HCO_3^-} + c^*_{CO_3^-}$$

$$A_T = c^*_{OH^-} + c^*_{HCO_3^-} + 2c^*_{CO_3^{2-}} + c^*_{CH_3COO^-} - c^*_{H^+}$$

$$C_{Na} = c^*_{Na^+}$$

$$C_{Ac} = c^*_{CH_3COO^-}$$

see appendix for derivation. Sodium and acetate does not participate in any equilibrium reactions and are therefore passive ions and remain as their provisional components. This leaves 5 unkonws $c^*_{H^+}$, $c^*_{OH^-}$, $c^*_{CO_2}$, $c^*_{HCO_3^-}$, $c^*_{CO_3^{2-}}$ and therby needs 5 corresponding equations to solve. The first is the carbon balance given by C_T, three are given by the reaction equilibriums

$$K_W = c_{H^+} c_{OH^-} \tag{20}$$

$$K_1 = \frac{c_{H^+} c_{HCO_3^-}}{c_{CO_2}} \tag{21}$$

$$K_2 = \frac{c_{H^+} c_{CO_3^{2-}}}{c_{HCO_3^-}} \tag{22}$$

And the last from electro-neutrality which can be expressed via the sodium and alkalinity components

$$\sum_i z_i c_i = 0 \Rightarrow C_{Na} - A_T = 0 \tag{23}$$

A convenience of this system is that all 5 equations can be expressed through $h = H^+$

$$c_{OH^-} = \frac{K_W}{h} \tag{24}$$

$$c_{CO_2} = C_T \frac{h^2}{D} \tag{25}$$

$$c_{HCO_3^-} = C_T \frac{K_1 h}{D} \tag{26}$$

$$c_{CO_3^{2-}} = C_T \frac{K1K2}{D} \tag{27}$$

$$C_{Na} - A_T = c_{Na^+} + h - \frac{K_W}{h} - C_T\frac{K_1 h}{D} - 2C_T\frac{K1K2}{D} - c_{CH_3COO^-} = 0 \tag{28}$$

using

$$D = h^2 + K_1 h + K_1 K_2 \tag{29}$$

The final equation becomes a constraint equation dependent only on h

$$f(h) = c_{Na^+} + h - \frac{K_W}{h} - C_T\frac{K_1 h}{D} - 2C_T\frac{K1K2}{D} - c_{CH_3COO^-} = 0 \tag{30}$$

see appendix for derivation. With this $H^{+(k+1)}$ can be solved iteratively through Newtons or Brent's method see appendix. Then use the updated $H^{+(k+1)}$ to update the other species via equations 24-27

$$c_{H^+}^{(k+1)},\ c_{OH^-}^{(k+1)},\ c_{CO_2}^{(k+1)},\ c_{HCO_3^-}^{(k+1)},\ c_{CO_3^{2-}}^{(k+1)}$$

Then the potential is updated based on the projected equilibrium concentrations through current conservation

$$\nabla \cdot j = 0 \tag{31}$$

with

$$j = -F\sum_i z_i D_i \nabla c_i^{(k+1)} - \frac{F^2}{RT}\sum_i z_i^2 D_i c_i^{(k+1)} \nabla \phi^{(k+1)} \tag{32}$$

The updated potential are relaxed

$$\phi^{(k+1)} = \alpha_\phi \tilde{\phi}^{(k+1)} + (1-\alpha_\phi)\phi^{(k)} \tag{33}$$

where $\tilde{\phi}^{(k+1)}$ is the values before relaxation. Iterate for fixed steps or until converged to acceptable residuals. Then update pH

$$pH = -log_{10}\left(\frac{c_{H^+}^{(k+1)}}{1000}\right) \tag{34}$$

convert concentration fields into mass-fraction

$$Y_i^{(k+1)} = c_i^{(k+1)}\frac{M_i}{\rho} \tag{35}$$

and perform inert species closure

$$Y_j^{(k+1)} = 1 - \sum_{i \neq j} Y_i^{(k+1)} \tag{36}$$

2.1 Algorithm flowchart

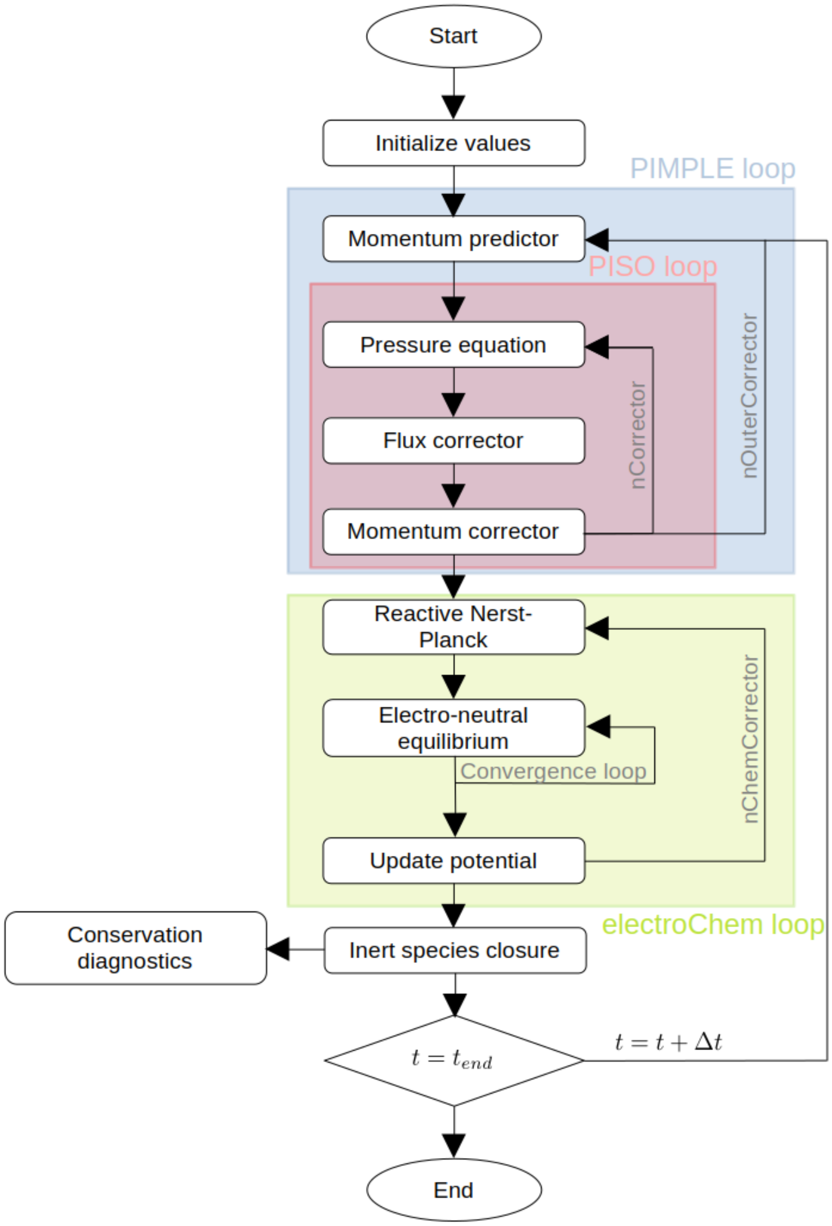

Figure 2: Enter Caption

3 Appendix

3.1 Conserved component derivation

The fast equilibrium reactions are separated from the kinetic electrode and microbial reactions. For the reduced water-carbonate system, the equilibrium reactions are

$$H_2O \leftrightarrow H^+ + OH^- \tag{37}$$

$$CO_2 + H_2O \leftrightarrow HCO_3^- + H^+ \tag{38}$$

$$HCO_3^- \leftrightarrow CO_3^{2-} + H^+ \tag{39}$$

Water is treated as inert and is therefore omitted from the vector of dissolved species. The aqueous species vector is defined as

$$\mathbf{c} = \begin{bmatrix} c_{H^+} & c_{OH^-} & c_{CO_2} & c_{HCO_3^-} & c_{CO_3^{2-}} & c_{Na^+} & c_{CH_3COO^-} \end{bmatrix}^T \tag{40}$$

Using positive stoichiometric coefficients for products and negative coefficients for reactants, the equilibrium stoichiometric matrix becomes

$$S_e = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \tag{41}$$

The equilibrium source terms can be represented as

$$\mathbf{R}_{eq} = S_e^T \dot{\boldsymbol{\xi}}_{eq} \tag{42}$$

where $\dot{\boldsymbol{\xi}}_{eq}$ contains the rates of the equilibrium reactions. The equilibrium rates are not calculated explicitly because the reactions are assumed instantaneous.

A conserved component is a linear combination of species that is unaffected by all equilibrium reactions. Defining the component matrix U, this requires

$$US_e^T = 0 \tag{43}$$

Multiplication of the equilibrium source term by U then gives

$$U\mathbf{R}_{eq} = US_e^T \dot{\boldsymbol{\xi}}_{eq} = 0 \tag{44}$$

The equilibrium reactions therefore do not appear as source terms in the transport equations of the components. There are seven aqueous species and three independent equilibrium reactions. The number of independent conserved components is therefore

$$N_C = N_S - \text{rank}(S_e) = 7 - 3 = 4 \tag{45}$$

A physically convenient basis for the null space is

$$U = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{46}$$

It can be verified directly that the matrix satisfies Eq. (43). The corresponding component vector is

$$\mathbf{C} = U\mathbf{c} = \begin{bmatrix} C_T \\ A_T \\ C_{Na} \\ C_{Ac} \end{bmatrix} \tag{47}$$

where the components are defined below.

$$C_T = c_{CO_2} + c_{HCO_3^-} + c_{CO_3^-}$$

$$A_T = c_{OH^-} + c_{HCO_3^-} + 2c_{CO_3^{2-}} + c_{CH_3COO^-} - c_{H^+}$$

$$C_{Na} = c_{Na^+}$$

$$C_{Ac} = c_{CH_3COO^-}$$

3.2 Reduction of the equilibrium problem

Starting from

$$C_T = c_{CO_2} + c_{HCO_3^-} + c_{CO_3^{2-}}$$
$$K_W = hc_{OH^-}$$
$$K_1 = \frac{hc_{HCO_3^-}}{c_{CO_2}}$$
$$K_2 = \frac{hc_{CO_3^{2-}}}{c_{HCO_3^-}}$$

and

$$\sum_i z_i c_i = C_{Na} - A_T = c_{Na^+} + h - c_{OH^-} - c_{HCO_3^-} - 2c_{CO_3^{2-}} - c_{CH_3COO^-} = 0$$

From this the equilibrium equations determine the ratio of each carbonate species and the mass balance gives the absolute concentration. From water dissociation

$$K_W = hc_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{h} \tag{48}$$

From the first carbonate equilibrium

$$K_1 = \frac{hc_{HCO_3^-}}{c_{CO_2}} \Rightarrow c_{HCO_3^-} = \frac{K_1}{h} c_{CO_2} \tag{49}$$

From the second carbonate equilibrium

$$K_2 = \frac{hc_{CO_3^{2-}}}{c_{HCO_3^-}} \Rightarrow c_{CO_3^{2-}} = \frac{K_2}{h} c_{HCO_3^-} = \frac{K_1 K_2}{h^2} c_{CO_2} \tag{50}$$

Now substitute these expressions into the carbonate mass balance

$$C_T = c_{CO_2} + \frac{K_1}{h} c_{CO_2} + \frac{K_1 K_2}{h^2} c_{CO_2} \tag{51}$$

factor out c_{CO_2}

$$C_T = c_{CO_2} \left(1 + \frac{K_1}{h} + \frac{K_1 K_2}{h^2}\right) \tag{52}$$

multiply the numerator and denominator by h^2

$$C_T = c_{CO_2} \left(\frac{h^2 + hK_1 + K_1 K_2}{h^2}\right) \tag{53}$$

Now define the numerator as D

$$D = h^2 + hK_1 + K_1 K_2 \tag{54}$$

Then

$$c_{CO_2} = C_T \frac{h^2}{D} \tag{55}$$
$$c_{HCO_3^-} = \frac{K_1}{h} c_{CO_2} = C_T \frac{K_1 h}{D} \tag{56}$$
$$c_{CO_3^{2-}} = \frac{K_1 K_2}{h^2} c_{CO_2} = C_T \frac{K_1 K_2}{D} \tag{57}$$

Finally substituting the electro-neutrality expression in terms of h gives the constraint equation

$$f(h) = c_{Na^+} + h - \frac{K_W}{h} - C_T \frac{K_1 h}{D} - 2C_T \frac{K_1 K_2}{D} - c_{CH_3COO^-} = 0 \tag{58}$$

3.3 Hybrid Newtons Brents method

c_{H^+} is solved through a constraint equation

$$f(h) = c_{Na^+} + h - \frac{K_W}{h} - C_T \frac{K_1 h}{D} - 2C_T \frac{K_1 K_2}{D} - c_{CH_3COO^-} = 0 \tag{59}$$

The derivitive of which is

$$f'(h) = \frac{d}{dh} f(h) = 1 + \frac{K_W}{h^2} + C_T K_1 \frac{h^2 + 4K_2 h + K_1 K_2}{D^2} \tag{60}$$

For positive concentrations and positive equilibrium constants,

$$f'(h) > 0 \tag{61}$$

The residual is therefore strictly increasing with h, so at most one positive solution exists. This makes the local problem well suited to a safeguarded Newton or Brent root-finding method. As Newtons method usually faster but potentially unstable than the more stable but slower Brent method the solver uses a hybrid Newton-Brent method. First Newtons method is tried for starting from the hydrogen-ion concentration from the previous iteration, $h^{(k)}$, Newton's method gives

$$h^{(k+1)} = h^{(k)} - \frac{f(h^{(k)})}{f'(h^{(k)})} \tag{62}$$

If the solution is not within n_{Newton} steps the solver switches to the Brent method with defined upper bound h_a and lower bound h_b such that $f(h_a) > 0$ and $f(h_b) < 0$. Then bisection is performed as

$$m = \frac{h_a + h_b}{2} \tag{63}$$

where $f(m)$ is evaluated. if $f(a)f(m) < 0$ the root is to the left else it is to the right. This is repeated using inverse quadratic interpolation